Iscandar vs Buddy under a Gaussian load

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Load: Gaussian Two (author's profile, 2026-10-26)
  request sizes ~ Normal(mu 512, sigma 170.5),
  rounded and clamped to 1..1023 sectors; the
  same seeded stream drives both engines.
Gaussian Two is this report's typical-load
demonstration: the runtime shape an RTOS
application presents to its allocator.

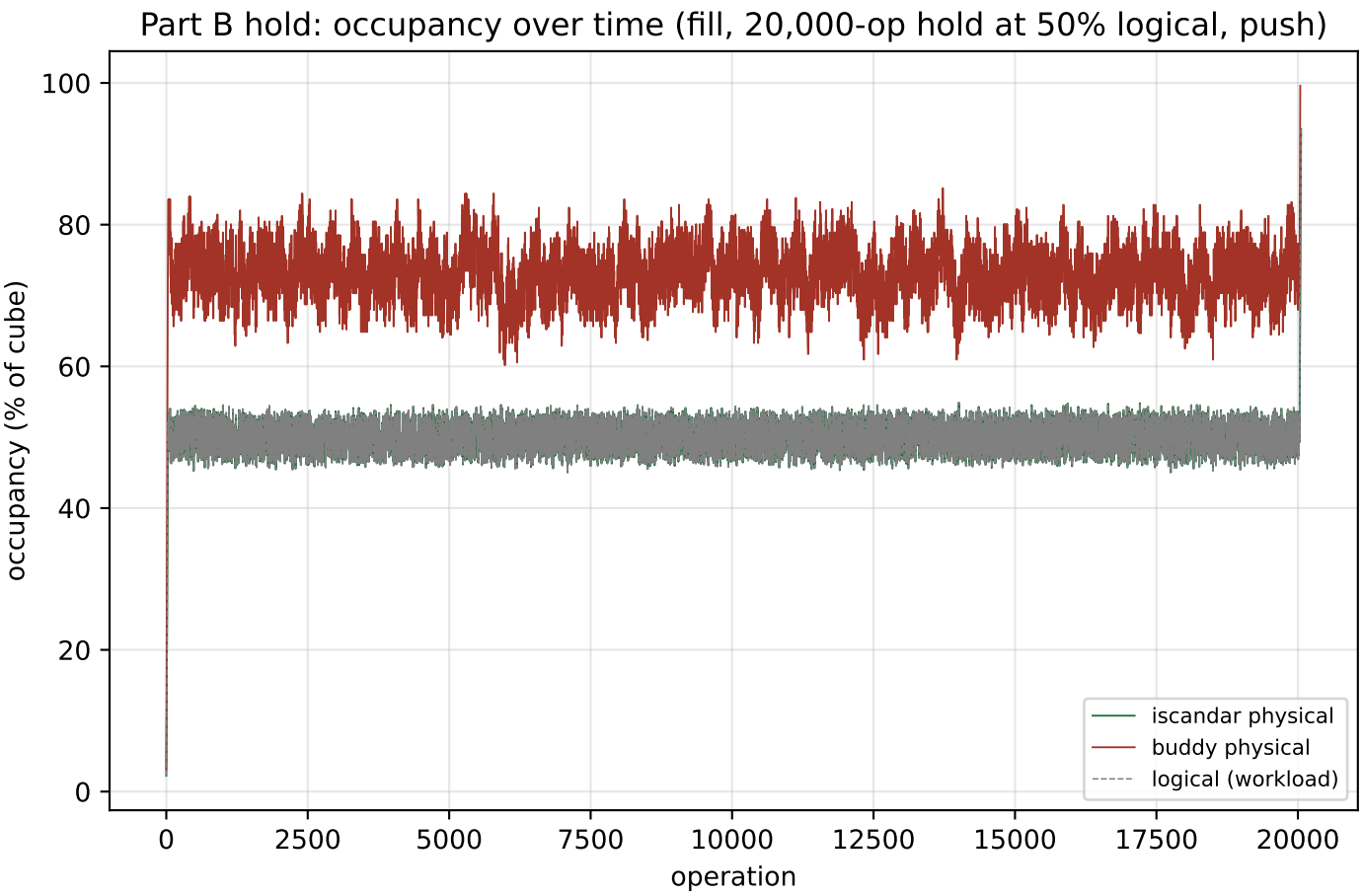
Engines: the full Iscandar engine (all modes,
  exact fit) vs the classic binary buddy.
Arena: 32,768 sectors (128 KiB at 4 B/sector).

Part A - the sprint (fill to 85% logical, churn,
push; first failed allocation loses):
  buddy    stalled op 38 in the ramp at 99.22%
           physical (iscandar then at 63.58%)
  iscandar stalled op 73 in the churn at 96.54%
           physical; margin 35 ops to iscandar.

Part B - the steady-state hold (fill to 50%
logical, 20,000 ops in a +/-2-point band, push):
  refusals during the hold: iscandar 0, buddy 0
  push stall: buddy at 99.61% physical, largest
           free block 128 sectors
  push stall: iscandar at 93.48% physical, largest
           free run 755 sectors

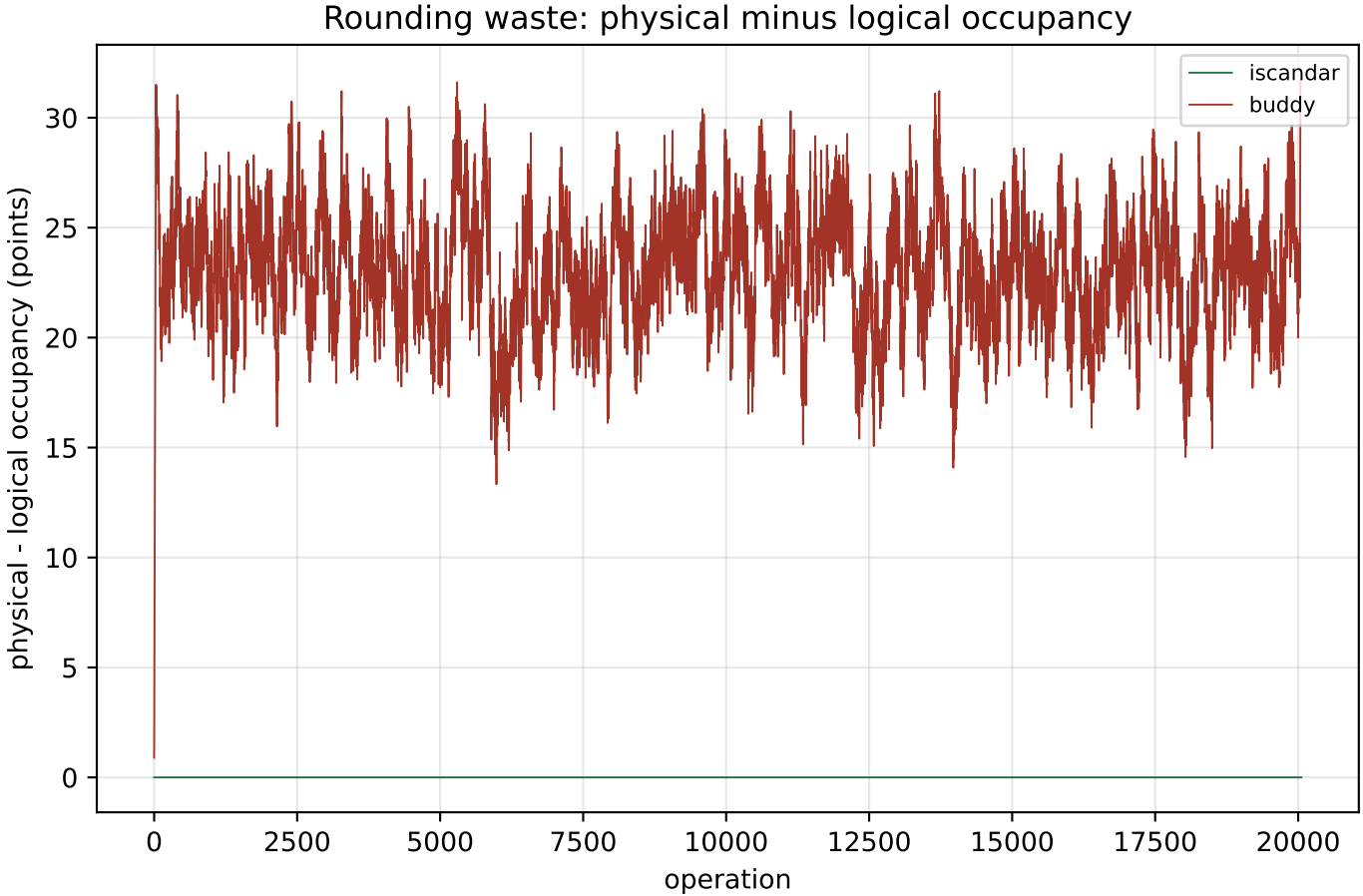
GammaCheck green in both parts (block checks
every op, full-ladder checks at boundaries).

Gaussian One (mu 16386, sigma 5461, full range)
is the envelope demonstrator, not an expected
load: its two-draw sum is the whole cube, so any
allocator fails the second concurrent request.
Its 2-op run stands in gaussian-1/ as the
capacity exhibit; the graphs here are Two's.
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Part B hold: occupancy over time (fill, 20,000-op hold at 50% logical, push)
occupancy (% of cube)
operation
0
20
40
60
80
100
0
2500
5000
7500
10000
12500
15000
17500
20000
iscandar physical
buddy physical
logical (workload)

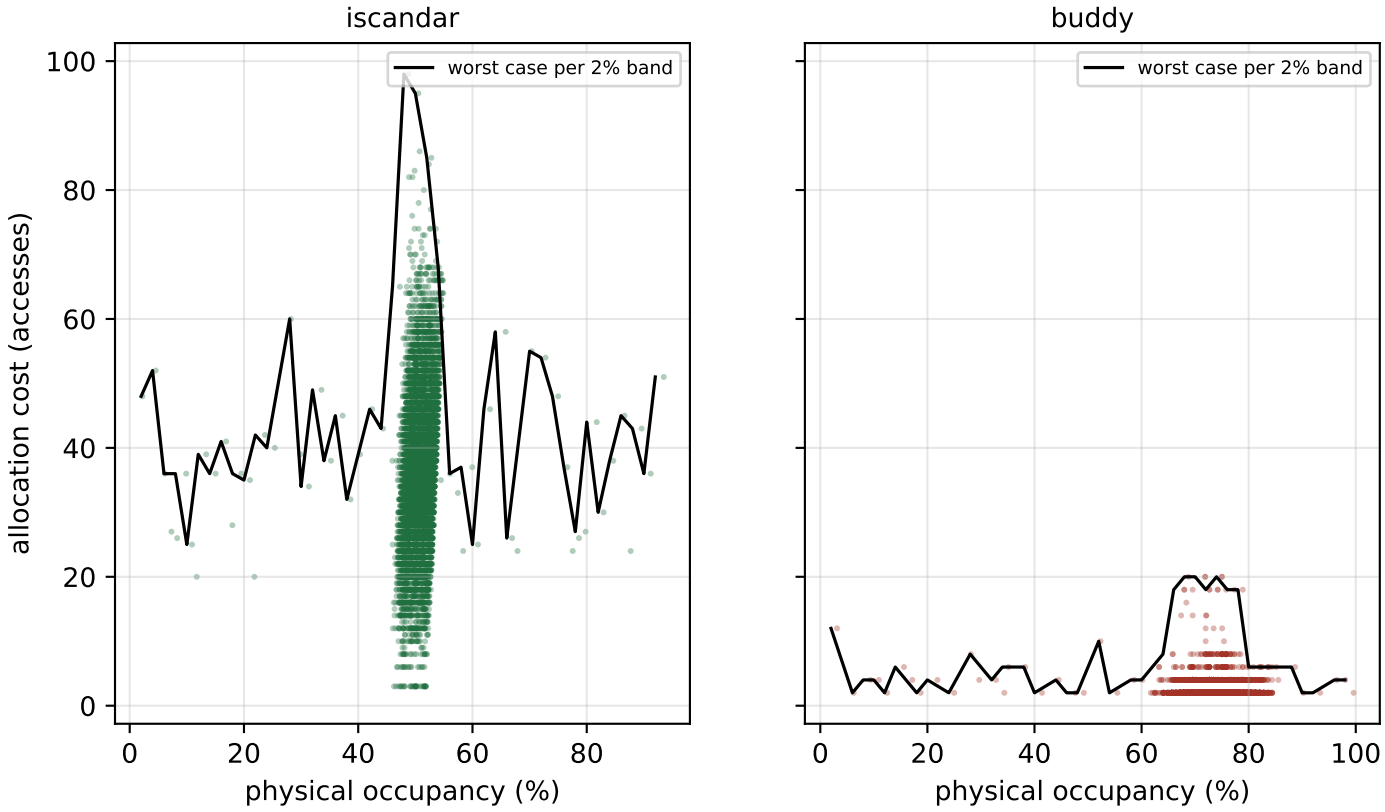
Commentary - occupancy over time

Three lines. The gray dashed line is the workload itself, held flat at 50% logical for 20,000 ops. Iscandar's green line lies exactly on top of it - it is barely visible separately, because exact fit means its physical occupancy is the logical occupancy, 45 to 55% and nothing more. Buddy's red line rides far above the whole time: it averages 73% physical and wanders between 60 and 85%. Same workload, same requests - Buddy's cube is three-quarters full while Iscandar's is half full.


Rounding waste: physical minus logical occupancy
iscandar
buddy
physical - logical occupancy (points)
0
5
10
15
20
25
30
0
2500
5000
7500
10000
12500
15000
17500
20000
operation

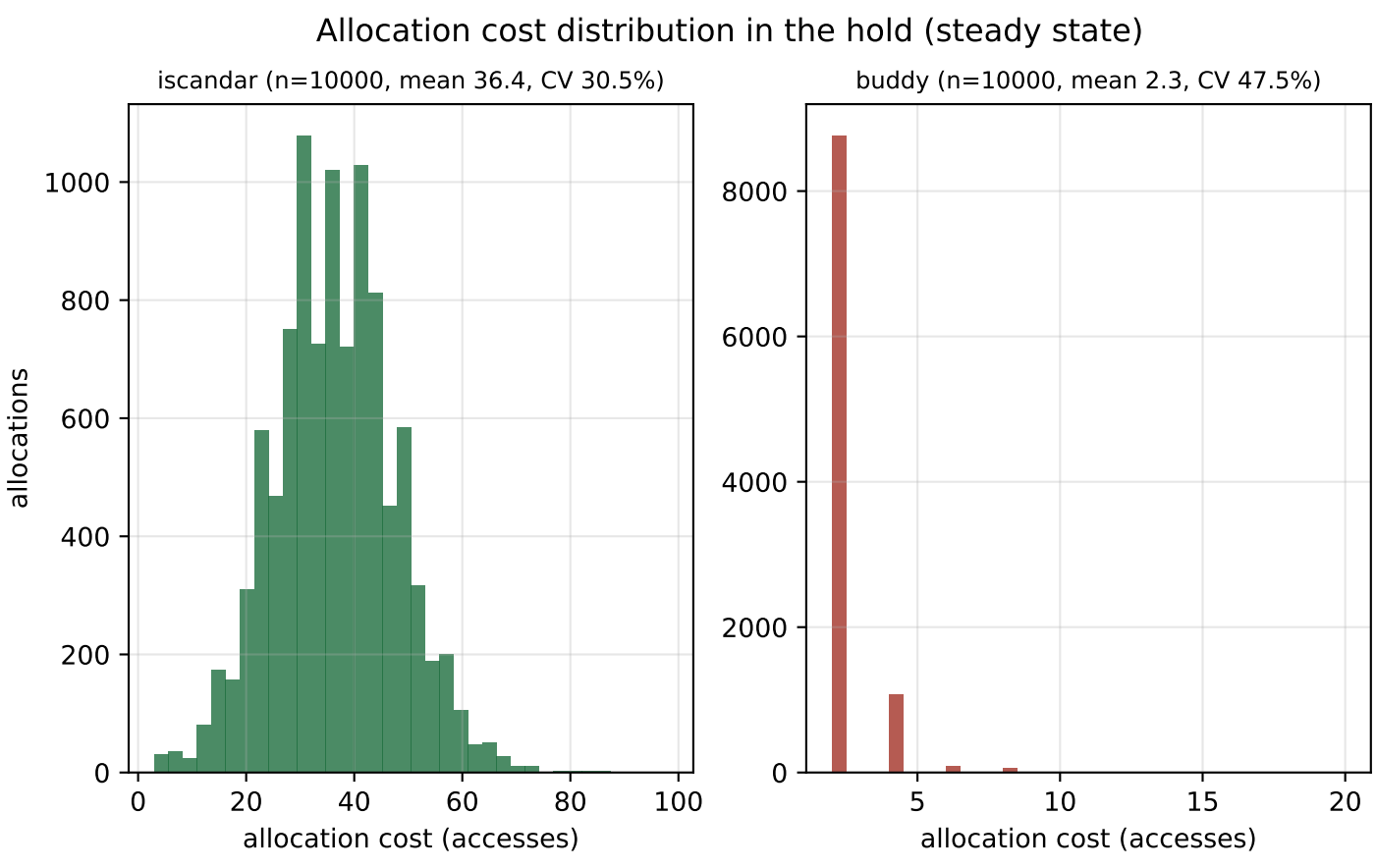
Commentary - rounding waste

This graph is the gap between the two engines drawn on its own. Iscandar is a flat line on zero for all 20,000 ops. Buddy is a noisy band around 23 points of pure waste, peaking at 31.6 - at its worst moments, nearly a third of Buddy's cube is neither data nor usable space, just rounding.


Allocation cost against occupancy (hold + push; units differ per engine, see closing page)
iscandar
buddy
worst case per 2% band
worst case per 2% band
allocation cost (accesses)
physical occupancy (%)
physical occupancy (%)
0
20
40
60
80
100

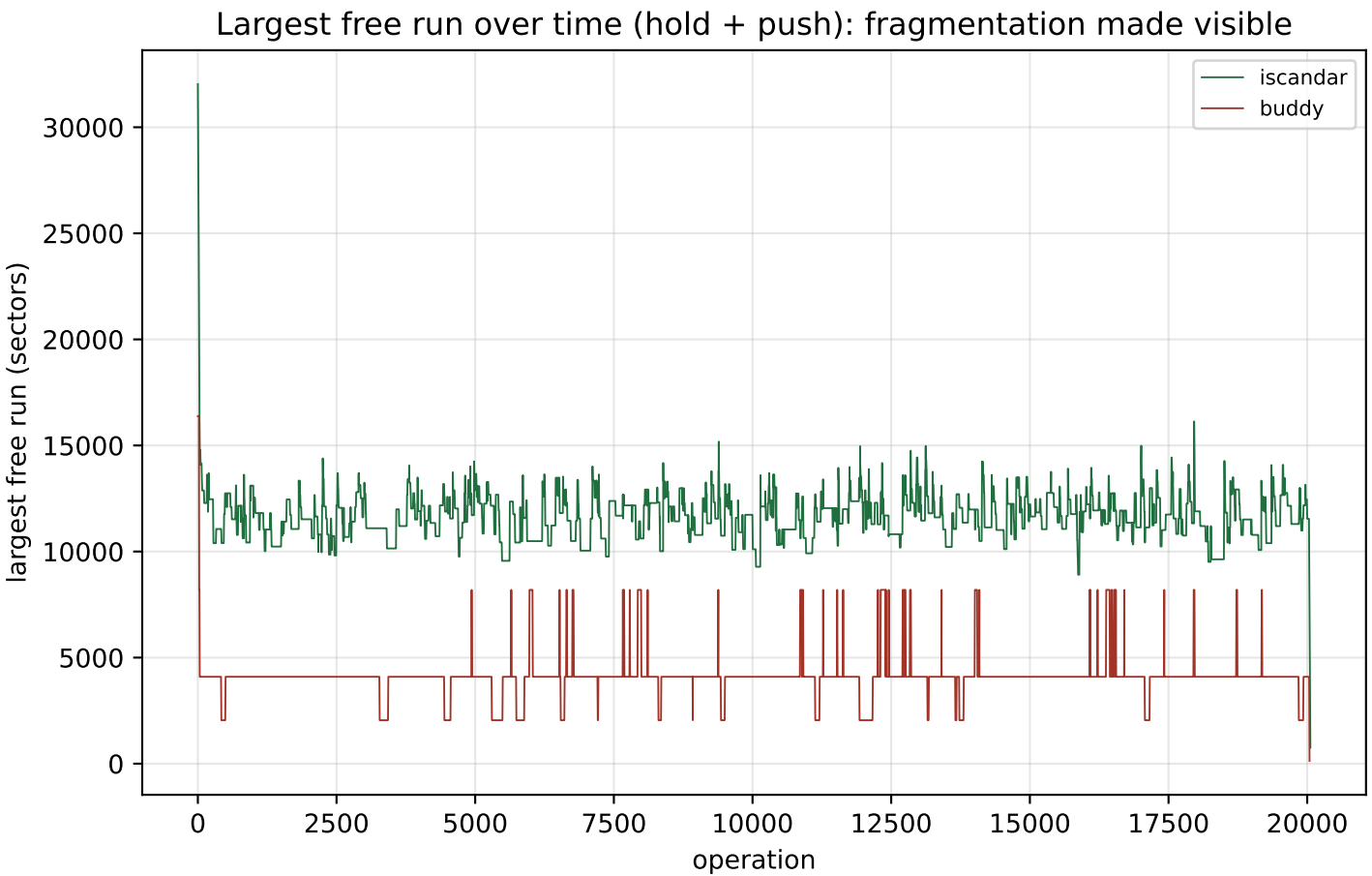
Commentary - cost against occupancy

Two panels with a black worst-case envelope line. On Iscandar's side, the dots form a cloud between about 20 and 100 touches, and the envelope stays between 45 and 98 across every occupancy band - it does not blow up as the cube fills; even in the 90-100% band the worst allocation costs 51. On Buddy's side, the dots hug the floor (its units are tree visits, so the levels are not comparable), with occasional spikes to 20 when the tree walk goes deep. The readable point: Iscandar's price grows gently and stays bounded; there is no cliff.


Allocation cost distribution in the hold (steady state)
iscandar (n=10000, mean 36.4, CV 30.5%)
buddy (n=10000, mean 2.3, CV 47.5%)
allocations
allocation cost (accesses)
allocation cost (accesses)
0
200
400
600
800
1000
0
2000
4000
6000
8000
0
20
40
60
80
100
5
10
15
20

Commentary - cost distributions

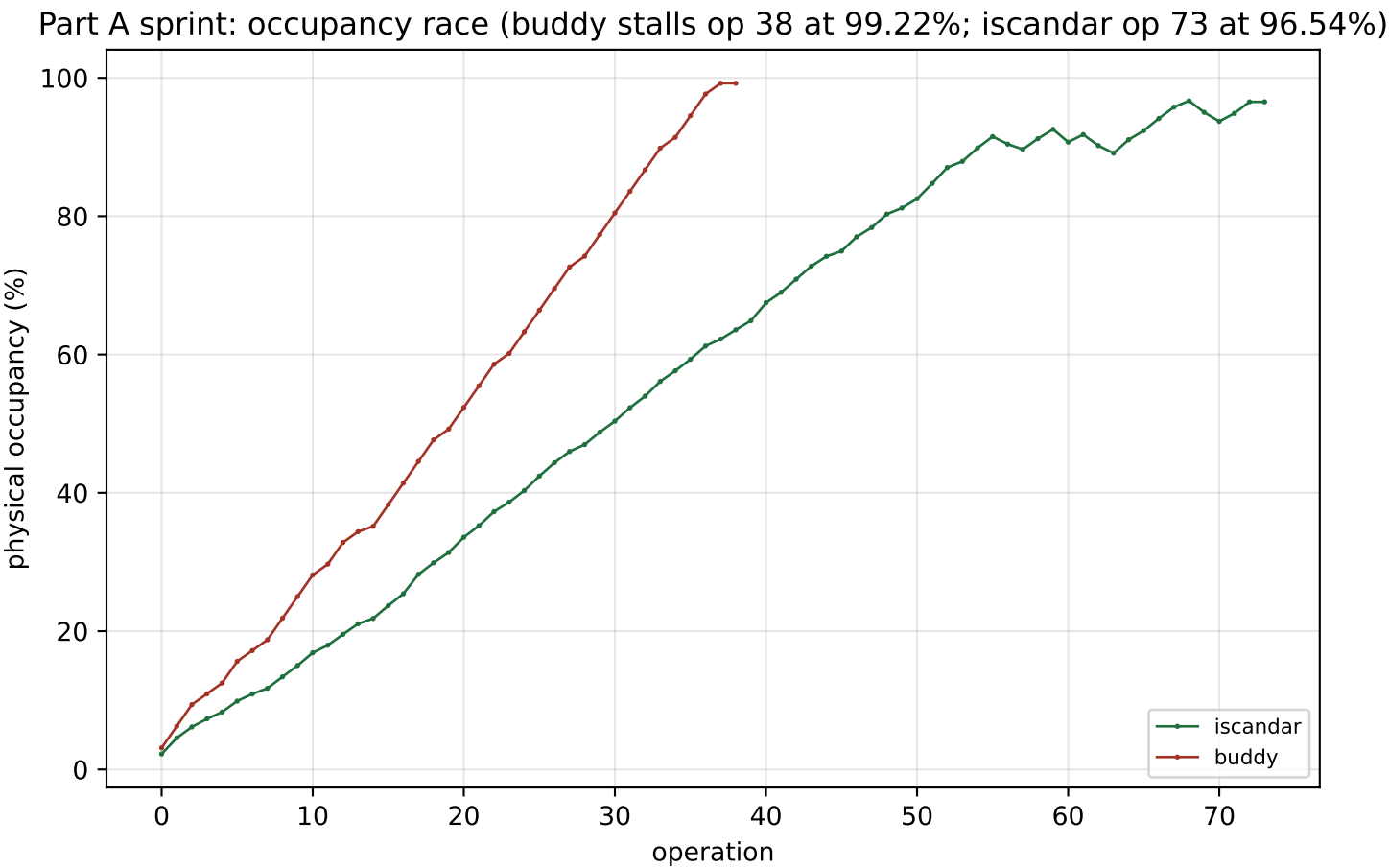
Iscandar's histogram is a mound centered at 36, with the middle half of allocations between 29 and 44 and the 90th percentile at 50. Buddy's is a spike at 2 - three quarters of its allocations cost the minimum - with a thin tail stretching to about 20. Different shapes entirely: a controlled mound against a spike-and-tail.


Largest free run over time (hold + push): fragmentation made visible
iscandar
buddy
largest free run (sectors)
operation
0
5000
10000
15000
20000
25000
30000
0
2500
5000
7500
10000
12500
15000
17500
20000

Commentary - largest free run

This is the fragmentation picture, and it is the most dramatic of the set. Iscandar's line rides high all through the hold: half the time its largest contiguous free run is 11,500 sectors or more, in a cube that is only half full. Buddy's line is a flat shelf at exactly 4,096 - its block lattice caps the largest hole it can ever offer at this occupancy - and at the end of the push it collapses to 128. Iscandar's line holds its altitude until the push, then falls to 755 at the stall. All hold long, Iscandar's biggest usable hole is roughly three times Buddy's.

Graph 5 is the one that makes the RTOS applications engineer stop and look. The occupancy graphs show what Buddy wastes; the largest-free-run graph shows what that waste costs - the size of the biggest promise each allocator can still keep.


Part A sprint: occupancy race (buddy stalls op 38 at 99.22%; iscandar op 73 at 96.54%)
physical occupancy (%)
operation
0
20
40
60
80
100
0
10
20
30
40
50
60
70
iscandar
buddy

Commentary - the sprint

Both lines climb from zero, but Buddy's climbs steeper from the very first operations - by op 2 it is at 9.4% against Iscandar's 6.1% on identical requests - and its line simply ends at op 38, at 99.22%. Iscandar's line keeps going alone to op 73, ending at 96.54%.

Numbers and caveats

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Numbers (population statistics over successful ops)

sprint iscandar alloc: n=66 mean 36.71 var 77.63 CV 24.0% stability 4.2 sigma, range 19..60

sprint buddy alloc: n=38 mean 4.00 var 5.68 CV 59.6% stability 1.7 sigma, range 2..12

hold   iscandar alloc: n=10000 mean 36.43 var 123.51 CV 30.5% stability 3.3 sigma, range 3..98
hold   iscandar free:  n=10000 mean 18.75 var 28.36 CV 28.4% stability 3.5 sigma, range 2..35

hold   buddy alloc: n=10000 mean 2.32 var 1.22 CV 47.5% stability 2.1 sigma, range 2..20
hold   buddy free:  n=10000 mean 2.16 var 0.30 CV 25.6% stability 3.9 sigma, range 2..11

Counting and caveats

- Iscandar accesses are sector-word touches on
  the engine path: probe words in the search plus
  the commit's word touches, per the unified
  model in gaussian_race.py's header. Gate
  (counter) reads are a separate series and are
  not blended into accesses.
- Buddy accesses are its tree-work analog units
  (buddy.py), NOT the same physical units. Read
  across engines by shape, CV, and outcome;
  read levels and sigmas only within one engine
  against its own bound (the two-series
  discipline adopted 2026-10-26).
- Stability sigma is mean / SD (the reciprocal
  of CV): how many of its own standard
  deviations the mean cost sits above zero. It
  is a stability ratio, not a defect-rate
  sigma; a defect sigma needs a declared limit
  (e.g. the Huge construction bound of 70
  accesses at size 1025, which no sample can
  exceed, so its defect rate is zero by
  construction and no finite sigma states it).
- Placement completeness (post-amendment): an
  Iscandar failure means no contiguous free run
  of the request existed, or the request
  exceeded all free space. It never fails facing
  an exact-fit hole.
- One seeded stream drove both engines in each
  part (seed 1). GammaCheck green throughout.
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